

INTERACTIVE PROBLEM SOLVER

Engineering Statics Problem Hub

An interactive solver for problems from Meriam & Kraige — Engineering Mechanics: Statics. Input your own values, watch the live diagram update, and explore step-by-step solutions.

2

Problems Solved

∞

Parametric Inputs



Fullscreen Mode

AVAILABLE SOLVERS

2/50

Chapter 2 · Moments & Couples



Forearm Moment Analysis

Determine the combined moment of weights about elbow pivot **O** and the required biceps tension **T** for equilibrium. Includes a realistic anatomical arm diagram with biceps tendon, forearm bone, and sphere.

IM_O = 0

Biceps Tension T

Elbow angle θ

Live Diagram

OPEN SOLVER



4/22

Chapter 4 · Structures · Trusses



Loaded Truss — Method of Joints

Determine forces in members **AB**, **CG**, and **DE** of the loaded truss. Uses global equilibrium to find reactions, then analyses joints A, C, and E using method of joints with 45° geometry.

Method of Joints

AB · CG · DE

45° roller at E

Live Diagram

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OPEN SOLVER



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Method of Joints

AB · CG · DE

45° roller at E

Live diagram

OPEN SOLVER



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Chapter 3 · Equilibrium

COMING SOON



3D Particle Equilibrium

Three-dimensional concurrent force systems, direction cosines, and equilibrium conditions for a particle in space.

3D Forces

Direction cosines

COMING SOON



MODULE 1 · MOMENTS & COUPLES · PROBLEM 2/50

Forearm Moment Analysis

Determine the combined moment about the elbow pivot O and find the required biceps tension T for equilibrium.

Problem 2/50 — Mass of forearm = 2.3 kg, CG at G . Find the combined moment of the weights about the elbow pivot O , and the biceps tension T so that $\Sigma M_O = 0$.

• LIVE DIAGRAM — UPDATES AS YOU TYPE

INPUT VALUES

GEOMETRY

θ — Elbow angle; upper arm to forearm ($^\circ$)

55

d_T — Horiz. distance O to biceps (mm)

50

d_G — Distance along arm O to G (mm)

150

d_A — Horiz. distance O to sphere A (mm)

325

MASSSES

m_{arm} — Forearm mass (kg)

2.3

m_s — Sphere mass (kg)

3.6

CONSTANT

g — Gravity (m/s^2)

9.81

⇒ COMPUTE SOLUTION

✓ Solution — Problem 2/50

[View Blueprint](#)

COMBINED MOMENT OF WEIGHTS

14.25 N·m

about O (Clockwise)

REQUIRED BICEPS TENSION T

285.0 Nfor $\Sigma M_O = 0$

STEP-BY-STEP WORKING

1 Weight of the forearm

$$\begin{aligned} &= \text{forearm mass} \times \text{gravitational acceleration} \\ &= 2.3 \times 9.81 = 22.563 \text{ N} \end{aligned}$$

Weight of the sphere

$$\begin{aligned} &= \text{sphere mass} \times \text{gravitational acceleration} \\ &= 3.6 \times 9.81 = 35.316 \text{ N} \end{aligned}$$

2 The elbow angle $\theta = 55^\circ$ is measured from the upper arm (vertical) to the forearm.
The forearm lies at $90^\circ - 55^\circ = 35.0^\circ$ below horizontal.

For a vertical force, the perpendicular moment arm about O = the horizontal distance from O.

Perp. dist. — forearm weight

$$= dG \times \sin(\theta) = 150 \times \sin(55^\circ) = 122.873 \text{ mm} = 0.123 \text{ m}$$

Perp. dist. — sphere weight

$$= dA = 325 \text{ mm} = 0.325 \text{ m}$$

3 Moment of forearm weight about O → Clockwise $\curvearrowright$

$$\begin{aligned} &= W_f \times dG \cdot \sin(\theta) = 2.3 \times 9.81 \times 150 \times \sin(55^\circ) / 1000 \\ &= 2.772 \text{ N·m (Clockwise } \curvearrowright) \end{aligned}$$

Moment of sphere weight about O → Clockwise $\curvearrowright$

$$\begin{aligned} &= W_s \times dA = 3.6 \times 9.81 \times 325 / 1000 \\ &= 11.478 \text{ N·m (Clockwise } \curvearrowright) \end{aligned}$$



$$= 3.6 \times 9.81 = 35.316 \text{ N}$$

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Moment of sphere weight about O → Clockwise $\curvearrowright$

$$= W_s \times dA = 3.6 \times 9.81 \times 325 / 1000$$

$$= 11.478 \text{ N}\cdot\text{m} \text{ (Clockwise } \curvearrowright)$$

- 4 Combined moment of weights about O

$$= M_{Wf} + M_{Ws}$$

$$= (2.3 \times 9.81 \times \sin(55^\circ) \times 150 / 1000) + (3.6 \times 9.81 \times 325 / 1000)$$

$$= (22.563 \times 0.123) + (35.316 \times 0.325)$$

$$= 2.772 + 11.478$$

$$= 14.25 \text{ N}\cdot\text{m} \text{ (Clockwise } \curvearrowright)$$

- 5 For $\sum M_O = 0$, the biceps tension T must create an equal and opposite moment.

$T \times dT = \text{combined moment of weights}$

$$T \times (50 / 1000) = 14.25 \text{ N}\cdot\text{m}$$

$$T \times 0.050 = 14.25$$

$$T = 14.25 / 0.050$$

$$T = 285.0 \text{ N} \rightarrow \text{Upward } \uparrow$$

MODULE 2 · TRUSSES · PROBLEM 4/22

Loaded Truss — Method of Joints

Determine the forces in all 13 members of the loaded truss using the method of joints, solved joint-by-joint.

Problem 4/22 — Five top-chord nodes (A–E) spaced d apart, with bottom-chord nodes H, G, F at depth d . Applied loads: $L/2$ at A & E, L at B, C, D (all downward). Pin at A, 45°-roller at E. **All 13 member forces** computed via method of joints.

• LIVE DIAGRAM — UPDATES AS YOU TYPE

■ Truss member

■ Applied load

■ Dimensions (d)

■ Reactions

INPUT VALUES

GEOMETRY

d — Spacing (mm)

1000

APPLIED LOAD

L — Full panel load (N)

10000

INFO (READ-ONLY)

Support A: Pin (A_x , A_y)

Support E: Roller at 45°

Loads: $L/2$ @ A & E, L @ B, C, D

Angle AH = 45° ($d=d$ triangle)

⇒ COMPUTE SOLUTION

A_x →

20.00 kN

Horizontal reaction

A_y ↑

20.00 kN

Vertical reaction at A

E ↗ (45°)

28.28 kN

Roller reaction at E



TOP CHORD

AB

35.00 kN

Compression (C)

BC

35.00 kN

Compression (C)

CD

35.00 kN

Compression (C)

DE

35.00 kN

Compression (C)

BOTTOM CHORD

HG

20.00 kN

Tension (T)

GF

20.00 kN

Tension (T)

VERTICALS

BH

10.00 kN

Compression (C)

CG

0

Zero-force member

DF

10.00 kN

Compression (C)

DIAGONALS

AH

21.21 kN

Tension (T)

HC

7.07 kN

Compression (C)

CF

7.07 kN

Compression (C)

FE

21.21 kN

Tension (T)

STEP-BY-STEP WORKING

1 Global equilibrium - reactions

$$\sum M_A = 0: -10000d - 210000d - 310000d - 210000d + E \cdot (1/\sqrt{2}) \cdot 4d = 0$$

$$E = 2\sqrt{2} \cdot 10000 = 28.28 \text{ kN}$$

$$\sum F_x = 0: A_x = 210000 \quad \sum F_y = 0: A_y = 210000$$

2 Joint A → AH (T), AB (C)

$$\sum F_y: 210000 - 10000/2 - AH/\sqrt{2} = 0$$

$$AH = 3\sqrt{2} \cdot 10000/2 = 21.21 \text{ kN T}$$

$$\sum F_x: 210000 + AH/\sqrt{2} - AB = 0$$

$$AB = 210000/2 + 21.21 \cdot \sqrt{2} = 35.00 \text{ kN C}$$

**2** Joint A → AH (T), AB (C)

$$\sum F_y: 210000 - 10000/2 - AH/\sqrt{2} = 0$$

$$AH = 3\sqrt{2}10000/2 = 21.21 \text{ kN } T$$

$$\sum F_x: 210000 + AH/\sqrt{2} - AB = 0$$

$$AB = 710000/2 = 35.00 \text{ kN } C$$

3 Joint E → FE (T), DE (C) [symmetric with A]

$$FE = 3\sqrt{2}10000/2 = 21.21 \text{ kN } T$$

$$DE = 710000/2 = 35.00 \text{ kN } C$$

4 Joint B → BH (C), BC (C)

$$\sum F_y: -10000 - BH = 0 \rightarrow BH = 10.00 \text{ kN } C$$

$$\sum F_x: BC = AB \rightarrow BC = 35.00 \text{ kN } C$$

5 Joint D → DF (C), CD (C) [symmetric with B]

$$DF = 10.00 \text{ kN } C$$

$$CD = 35.00 \text{ kN } C$$

6 Joint H → HC (C), HG (T)

$$\sum F_y: 310000/2 - 10000 + HC/\sqrt{2} = 0$$

$$HC = 10000\sqrt{2}/2 = 7.07 \text{ kN } C$$

$$\sum F_x: -310000/2 - 10000/2 + HG = 0$$

$$HG = 210000 = 20.00 \text{ kN } T$$

7 Joint F → CF (C), GF (T) [symmetric with H]

$$CF = 10000\sqrt{2}/2 = 7.07 \text{ kN } C$$

$$GF = 210000 = 20.00 \text{ kN } T$$

8 Joint G / Joint C — verify CG

$$\text{At G: } HG (-T), GF (-T), CG (+)$$

$$\sum F_x: -210000 + 210000 = 0 \checkmark$$

$$\sum F_y: CG = 0 \rightarrow CG = 0 \text{ (Zero-force member)}$$



$$\sum F_x: BC = AB \rightarrow BC = 35.00 \text{ kN } C$$

5 Joint D → DF (C), CD (C) [symmetric with B]

$$DF = 10.00 \text{ kN } C$$

$$CD = 35.00 \text{ kN } C$$

6 Joint H → HC (C), HG (T)

$$\sum F_y: 310000/2 - 10000 + HC/\sqrt{2} = 0$$

$$HC = 10000\sqrt{2}/2 = 7.07 \text{ kN } C$$

$$\sum F_x: -310000/2 - 10000/2 + HG = 0$$

$$HG = 210000 = 20.00 \text{ kN } T$$

7 Joint F → CF (C), GF (T) [symmetric with H]

$$CF = 10000\sqrt{2}/2 = 7.07 \text{ kN } C$$

$$GF = 210000 = 20.00 \text{ kN } T$$

8 Joint G / Joint C — verify CG

At G: HG (→T), GF (→T), CG (→)

$$\sum F_x: -210000 + 210000 = 0 \checkmark$$

$$\sum F_y: CG = 0 \rightarrow CG = 0 \text{ (Zero-force member)}$$

Verify at C: HC and CF each supply 10000/2 upward

$$+ 10000/2 + 10000/2 = 10000 \text{ balances applied load } \checkmark$$

9 All 13 Members — Summary

Top chord (C): $AB=BC=CD=DE = 710000/2 = 35.00 \text{ kN}$

Bot chord (T): $HG=GF = 210000 = 20.00 \text{ kN}$

Verticals (C): $BH=DF = 10000 = 10.00 \text{ kN}$

Vertical (0): $CG = 0 \text{ (zero-force)}$

Diag out (T): $AH=FE = 3\sqrt{2}10000/2 = 21.21 \text{ kN}$

Diag in (C): $HC=CF = 10000\sqrt{2}/2 = 7.07 \text{ kN}$